

Source: docs/FUTURE_HARDWARE.md • Status: Verified & Synchronized

Status: FUTURE / OUT OF SCOPE FOR V1

This specification documents forward-looking architectural designs and communication protocols for an autonomous physical robotic solver rig planned for Cubyntra V4. No physical robotics hardware or drivers are part of the Cubyntra V1 release.

The envisioned physical robotic solver consists of a 6-axis mechanical fixture clamping the six center caps of a standard 3 x 3 x 3 Rubik's Cube, enabling simultaneous or sequential rotation of any face without manual human intervention.

```
|
|           Cubyntra Web Application
| (Kociemba Two-Phase Solver -> Solution Move Sequence Array)
+-----+
|
|           %
|           Web Serial / Web Bluetooth
|           %
|           %%%
+-----+
|
|           ESP32 Master Microcontroller
| - Real-time command buffer
| - Kinematic S-curve acceleration profiling
| - Step/Dir pulse generation (FastAccelStepper)
+-----+
|
|           %
|           Step & Direction Pulses
|           %
|           %%%
+-----+
|
|           6x TMC2209 Silent Stepper Motor Drivers
+-----+
|
|           %
|           Bipolar Stepper Currents
|           %
|           %%%
+-----+
|
|           6x NEMA-17 Bipolar Stepper Motors
| (Direct-Drive / Planetary Gear to 6 Face Grippers)
|
```

2. Microcontroller & Driver Hardware Architecture

2.1 Compute Unit: ESP32-S3

- Dual-Core Xtensa LX7 @ 240 MHz:
- Core 0: High-speed Web Serial / BLE communications and command parsing.
- Core 1: Hardware timer interrupts generating step pulses up to 100 kHz.
- Onboard Flash: 8 MB for offline move caching.

2.2 Motor Drivers: TMC2209

- StealthChop2 / SpreadCycle: Ultra-quiet operation and high torque output during rapid accelerations.
- Microstepping: 1/16 microstepping with interpolation to 1/256 for smooth positioning.
- StallGuard4: Sensorless stall detection used for automatic home calibration and physical jam detection.

3. Communication Protocol (Web Serial / BLE)

3.1 Packet Specification

Commands are transmitted as lightweight, deterministic JSON or binary packets over standard Web Serial at 115,200 baud.

JSON Representation:

```
{
  "seq": 104,
  "cmd": "EXECUTE_SOLVE",
  "moves": [
    { "face": "R", "turns": 1, "speed_rpm": 450 },
    { "face": "U", "turns": -1, "speed_rpm": 450 },
    { "face": "F", "turns": 2, "speed_rpm": 600 }
  ]
}
```

Compact Binary Protocol (4 bytes per move):

```
[ Byte 0: Face ID ('U' = 0x01, 'D' = 0x02, 'F' = 0x03, 'B' = 0x04, 'L' = 0x05, 'R' = 0x06) ]
[ Byte 1: Rotation Direction (0x01 = +90°, 0xFF = -90°, 0x02 = 180°) ]
[ Byte 2: Acceleration Profile ID ]
[ Byte 3: CRC8 Checksum ]
```

4. Kinematics & Mechanical Safety Considerations

4.1 S-Curve Acceleration Profiling

Step motors undergoing instantaneous velocity changes induce mechanical shock, leading to piece popping or corner twisting:

- Jerk-limited third-order polynomial S-curve velocity profiles:
$$j(t) = \frac{da(t)}{dt} = \text{constant}$$
- Maximum rotational velocity: 600 RPM (100ms per 90° turn).

4.2 Interlock & Collision Prevention

A physical Rubik's cube cannot execute perpendicular face turns simultaneously (e.g., R and U cannot rotate at the exact same instant without collision):

- The firmware scheduler enforces a minimum 15ms inter-move dwell time.
- Opposite face turns (e.g., R and L, or U and D) may be executed concurrently to optimize physical solve time.

4.3 Position Feedback & Error Recovery

Each stepper axis is equipped with an AS5600 12-bit magnetic rotary encoder on the output shaft:

- Confirms precise $\pm 0.5^\circ$ angular alignment at the completion of every move.
- In the event of a mechanical bind, motion halts instantly, reporting an emergency stop packet to the Cubyntra web client.